

Relative Stability: Gain Margin and Phase Margin

How much headroom is there before instability?

Ogata, *Modern Control Engineering*, Ch. 7.

In this tutorial you will:

- define gain margin and phase margin and read them with `margin`,
- relate phase margin to the equivalent damping ratio, and
- watch the margins shrink to zero as the loop gain rises.

Step through with **Ctrl+Enter**, or render a report with **publish**.

For a minimum-phase open-loop system whose Nyquist plot crosses the real axis to the right of -1 , two scalar measures of **relative** stability (margin before instability) are defined:

Gain margin $GM = \frac{1}{|G(j\omega_p)|}$ (often in dB: $GM_{dB} = -20 \log_{10} |G(j\omega_p)|$), evaluated at the **phase crossover frequency** ω_p where $\angle G(j\omega_p) = -180^\circ$ – the factor by which the gain can increase before instability.

Phase margin $PM = 180^\circ + \angle G(j\omega_c)$, evaluated at the **gain crossover frequency** ω_c where $|G(j\omega_c)| = 1$ – the additional phase lag that can be tolerated before instability.

Contents

- Worked Example
- Interpreting the Margins
- Phase Margin vs. Damping Ratio (Approximate Correlation)
- Sweeping Gain: Margins Shrink Toward Instability
- Before vs. After in the Time Domain
- Try it yourself

Worked Example

$G(s) = \frac{5}{s(s+1)(s+4)}$ (Gain 5 sits comfortably below the critical gain of 20 for this plant, so the margins come out positive – exactly the healthy case the next section interprets. Watch what happens to these numbers at $K = 20$ in the gain sweep further down.)

```
G = tf(5,conv([1 0],conv([1 1],[1 4])));

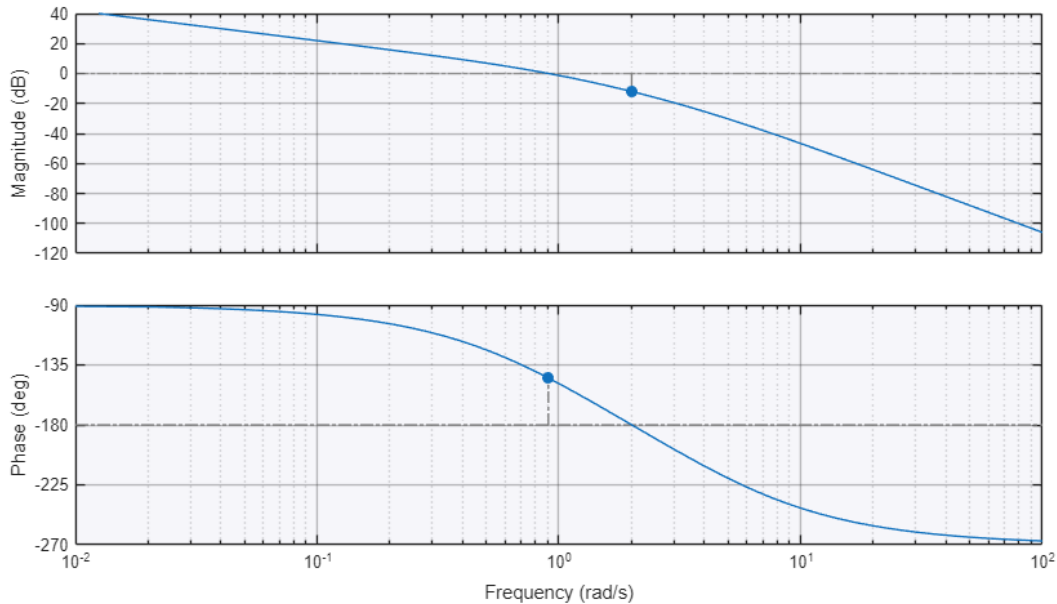
[Gm,Pm,Wcp,Wcg] = margin(G);
Gm dB = 20*log10(Gm);
fprintf('Gain margin = %.4f (%.2f dB) at wp = %.4f rad/s\n', Gm, Gm dB, Wcp)
fprintf('Phase margin = %.4f deg at wc = %.4f rad/s\n', Pm, Wcg)

figure
margin(G)
title('Bode Plot with Gain/Phase Margin','Interpreter','latex','FontSize',20)
grid on

Gain margin = 4.0000 (12.04 dB) at wp = 2.0000 rad/s
Phase margin = 35.1378 deg at wc = 0.9043 rad/s
```

Bode Plot with Gain/Phase Margin

Gm = 12 dB (at 2 rad/s), Pm = 35.1 deg (at 0.904 rad/s)



Interpreting the Margins

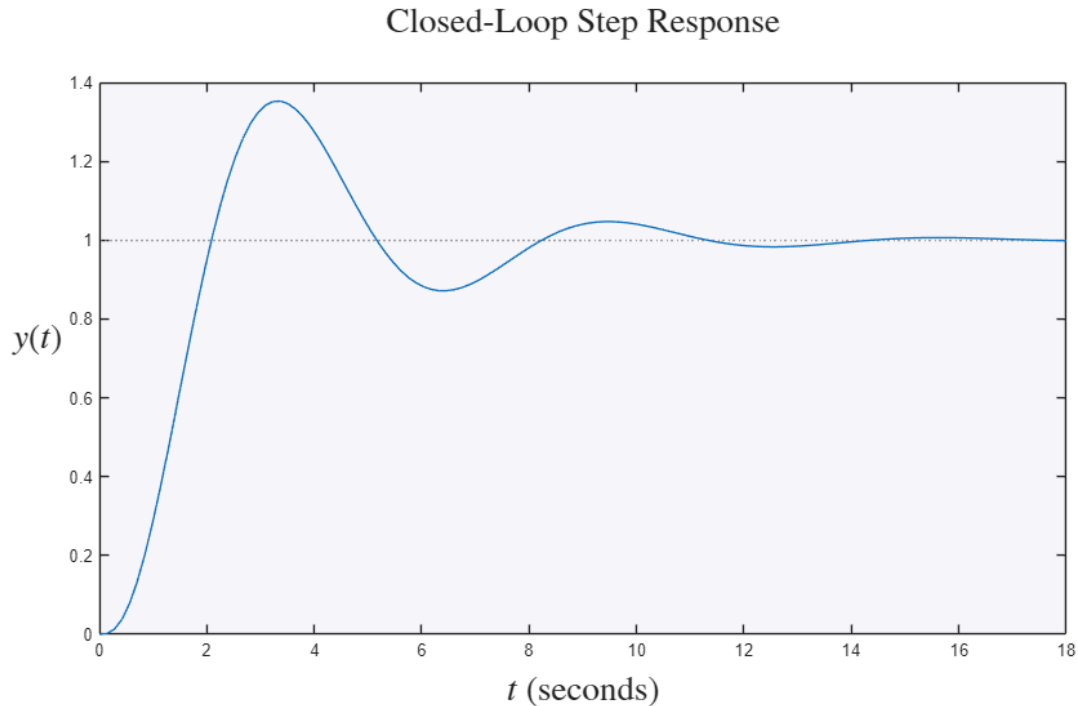
Both margins positive ($GM > 1$, i.e. $GM_{dB} > 0$, and $PM > 0^\circ$) indicates closed-loop stability; small positive margins indicate a lightly-damped, oscillatory closed-loop response, while large margins indicate sluggish but robust response. A practical rule of thumb (per Ogata) targets $PM \approx 30^\circ - 60^\circ$ and $GM_{dB} > 6$ dB.

```
T = feedback(G,1);
fprintf('Closed-loop poles:\n'); disp(pole(T))
fprintf('Closed-loop stable: %d\n', all(real(pole(T))<0))

figure
step(T)
title('Closed-Loop Step Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

```
Closed-loop poles:
-4.3442 + 0.0000i
-0.3279 + 1.0215i
-0.3279 - 1.0215i
```

```
Closed-loop stable: 1
```



Phase Margin vs. Damping Ratio (Approximate Correlation)

For a second-order-dominant system, a commonly used approximation relates phase margin (in degrees) to the equivalent damping ratio:

$$\zeta \approx \frac{PM}{100} \text{ (for } PM \text{ up to about } 60^\circ)$$

```
zeta_approx = Pm/100;
fprintf('Approximate equivalent damping ratio from PM: zeta = %.4f\n', zeta_approx)
```

```
Approximate equivalent damping ratio from PM: zeta = 0.3514
```

Sweeping Gain: Margins Shrink Toward Instability

As the loop gain rises the phase margin shrinks and eventually goes negative (closed-loop unstable). Print GM/PM across a range:

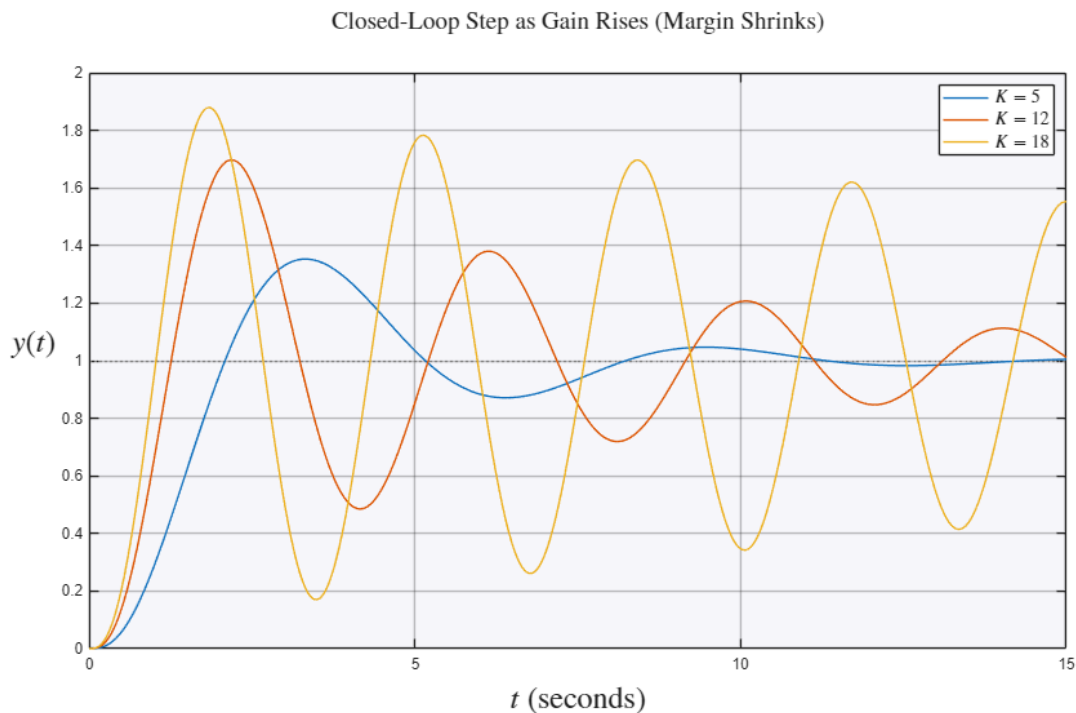
```
Ks = [5 20 60 96];
for k = Ks
    Gk = tf(k,conv([1 0],conv([1 1],[1 4])));
    [gm,pm] = margin(Gk);
    fprintf('K=%5.1f: GM=%.3f (%.2f dB), PM=%.2f deg\n', k, gm, 20*log10(gm), pm)
end
```

```
K= 5.0: GM=4.000 (12.04 dB), PM=35.14 deg
K= 20.0: GM=1.000 (-0.00 dB), PM=0.00 deg
Warning: The closed-loop system is unstable.
K= 60.0: GM=0.333 (-9.54 dB), PM=-22.98 deg
Warning: The closed-loop system is unstable.
K= 96.0: GM=0.208 (-13.62 dB), PM=-31.45 deg
```

Before vs. After in the Time Domain

The same trend as closed-loop step responses (using stable gains so the curves stay on screen): more gain $\rightarrow$ less phase margin $\rightarrow$ more overshoot and ringing, marching from "before" (well damped) toward instability.

```
Ks_stable = [5 12 18];
figure
hold on
for k = Ks_stable
    step(feedback(tf(k,conv([1 0],conv([1 1],[1 4]))),1), 0:0.02:15)
end
hold off
grid on
legend(arrayfun(@(k) sprintf('$K=%d$',k), Ks_stable,'UniformOutput',false), ...
    'Interpreter','latex','FontSize',12)
title('Closed-Loop Step as Gain Rises (Margin Shrinks)','Interpreter','latex','FontSize',16)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)
```



Try it yourself

- Push a stable gain from 5 toward 20 and watch `margin` report the phase margin shrinking toward zero as the step response rings harder.
- Use the approximation $\zeta \sim \text{PM}/100$ to predict overshoot, then check it against `stepinfo(feedback(Gk,1))`.